Convexity of a volume-sampling error sequence

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Verified affirmative resolution of RA-07. Theorem 1.1 and its proof; Sections 2–3 establish sampling corollaries. Independent agent verification is not external human peer review or formal certification. Original draft remarks about review or source access reflect the input date; the dated independent review records the subsequent checks.

Abstract

For positive numbers $\lambda_1, \dots, \lambda_n$, let e_j be their elementary symmetric polynomials, with $e_0 = 1$ and $e_{n+1} = 0$. We prove that $(j+1)e_{j+1}/e_j$ is a decreasing convex sequence on $\{0, \dots, n\}$. The proof reduces each second difference to the nonnegative quantity $s_1 s_3 - s_2^2$ for the reciprocal roots of a derivative of the generating polynomial. This proves Conjecture 1 of Dereziński, Khanna, and Mahoney. We give a self-contained application to fixed-size determinantal sampling, remove the concentration loss from its stable-rank bound, and derive a slightly sharper explicit bound.

1 The discrete inequality

For $0 \le j \le n$, write

$$e_j = \sum_{\substack{S \subseteq \{1,\dots,n\} \\ |S| = j}} \prod_{i \in S} \lambda_i, \qquad F_j = (j+1)\frac{e_{j+1}}{e_j}.$$

Here an empty product is one, and $e_{n+1} = 0$. In particular $F_n = 0$. The following theorem proves the convexity conjecture stated in [1, Section 5, Conjecture 1].

Theorem 1.1. *If every λ_i is positive, then $F_0 > F_1 > \cdots > F_n$ and*

$$F_{k-1} - 2F_k + F_{k+1} \ge 0 \qquad (1 \le k \le n-1).$$

More precisely, factor

$$Q(t) := \frac{d^{k-1}}{dt^{k-1}} \prod_{i=1}^{n} (1 + \lambda_i t) = Q(0) \prod_{a=1}^{n-k+1} (1 + \mu_a t), \qquad s_\ell = \sum_a \mu_a^\ell.$$

All μ_a are positive, and

$$F_{k-1} - 2F_k + F_{k+1} = \frac{2(s_1 s_3 - s_2^2)}{s_1(s_1^2 - s_2)} = \frac{2\sum_{a<b} \mu_a \mu_b (\mu_a - \mu_b)^2}{s_1(s_1^2 - s_2)}. \tag{1}$$

Proof. Put $P(t) = \prod_i (1 + \lambda_i t)$. Since $P^{(j)}(0) = j! e_j$,

$$F_j = \frac{P^{(j+1)}(0)}{P^{(j)}(0)}.$$

Every root of P is strictly negative. Repeated applications of Rolle's theorem, including the usual multiplicities at repeated roots, show that every derivative of positive degree also has only strictly negative roots. Thus Q has the stated factorization with positive μ_a.

The first three derivatives of this product at zero give

$$\frac{Q'(0)}{Q(0)} = s_1, \qquad \frac{Q''(0)}{Q(0)} = s_1^2 - s_2, \qquad \frac{Q'''(0)}{Q(0)} = s_1^3 - 3s_1 s_2 + 2s_3.$$

Consequently

$$F_{k-1} = s_1, \qquad F_k = \frac{s_1^2 - s_2}{s_1}, \qquad F_{k+1} = \frac{s_1^3 - 3s_1 s_2 + 2s_3}{s_1^2 - s_2}.$$

Subtracting gives the first identity in (1). Its denominator is positive because Q has degree at least two and $s_1^2 - s_2 = 2\sum_{a<b} \mu_a \mu_b > 0$. The numerator satisfies

$$s_1 s_3 - s_2^2 = \sum_{a<b} \mu_a \mu_b (\mu_a - \mu_b)^2 \geq 0.$$

This proves convexity, including $k = n - 1$: in that case Q has degree two and $Q''' = 0$.

For monotonicity, use the same calculation with $Q = P^{(j)}$, for $0 \leq j \leq n - 1$. It gives $F_j - F_{j+1} = s_2/s_1 > 0$. When Q has degree one, this identity still holds, with $F_{j+1} = F_n = 0$. $\square$

If all λ_i equal λ, then $F_j = (n - j)\lambda$ and every second difference vanishes. Formula (1) also supplies an explicit nonnegative certificate in the general case; no limiting or probabilistic argument is needed for the theorem.

2 Determinantal sampling

Let $A \in \mathbb{R}^{m \times N}$ have rank r, and let $\lambda_1 \geq \cdots \geq \lambda_r > 0$ be the nonzero eigenvalues of $G = A^T A$. Define

$$\mathcal{E}(S) = \|A - P_S A\|_F^2,$$

where P_S is the orthogonal projector onto the span of the columns indexed by S. For $0 \leq j \leq r$, the fixed-size determinantal distribution assigns probability proportional to $\det(G_{S,S})$ to subsets with $|S| = j$.

Lemma 2.1. *For fixed-size sampling of size j,*

$$\mathbb{E}_j \mathcal{E}(S) = F_j = (j+1) e_{j+1}/e_j.$$

For the random-size distribution

$$\mathbb{P}_\alpha(S) = \frac{\alpha^{-|S|} \det(G_{S,S})}{\det(I + G/\alpha)}, \qquad \alpha > 0,$$

write $K = |S|$. Then

$$\mu(\alpha) := \mathbb{E}_\alpha K = \sum_{i=1}^{r} \frac{\lambda_i}{\lambda_i + \alpha}, \qquad \mathbb{E}_\alpha \mathcal{E}(S) = \alpha \mu(\alpha). \tag{2}$$

Proof. The principal-minor identity gives $\sum_{|S|=j} \det(G_{S,S}) = e_j$. For $i \notin S$, the Gram determinant formula gives

$$\det(G_{S,S})\,\|(I-P_S)a_i\|_2^2 = \det(G_{S\cup\{i\},S\cup\{i\}}).$$

This also holds when $G_{S,S}$ is singular: adding one column cannot increase its rank to $j+1$, so both sides vanish. Summing over S and i counts every $(j+1)$-set exactly $j+1$ times, proving the fixed-size formula.

Put $P(t) = \sum_{j=0}^{r} e_j t^j = \prod_{i=1}^{r}(1+\lambda_i t)$. The probability generating function of K has normalizing factor $P(1/\alpha)$. Thus

$$\mathbb{E}_\alpha K = \frac{\alpha^{-1}P'(1/\alpha)}{P(1/\alpha)}, \qquad \mathbb{E}_\alpha \mathcal{E}(S) = \frac{P'(1/\alpha)}{P(1/\alpha)}.$$

The logarithmic derivative of P proves (2). $\square$

Corollary 2.2. *Let $1 \le k < r$. If $\mu(\alpha) \le k$, then*

$$\mathbb{E}_k\mathcal{E}(S) \le \alpha\mu(\alpha) \le \alpha k.$$

In particular, there is a unique $\alpha_k > 0$ satisfying $\mu(\alpha_k) = k$, and

$$\mathbb{E}_k\mathcal{E}(S) \le \mathbb{E}_{\alpha_k}\mathcal{E}(S) = \alpha_k k.$$

Proof. Let $F(x)$ be the piecewise-linear interpolation of the values F_j on $[0,r]$. Theorem 1.1 makes F convex and decreasing. Conditional on $K=j$, random-size sampling is the fixed-size distribution, so

$$F(k) \le F(\mathbb{E}K) \le \mathbb{E}F(K) = \mathbb{E}_\alpha\mathcal{E}(S).$$

The remaining assertions follow from (2) and the strict decrease of $\mu(\alpha)$ from r to zero as α increases from zero to infinity. $\square$

3 Stable-rank bounds without a concentration loss

For $0 \le s < r$, define the order-s stable rank by

$$r_s = \frac{\sum_{i>s}\lambda_i}{\lambda_{s+1}}, \qquad t_s = s + r_s.$$

The optimal rank-k squared Frobenius error is $T_k = \sum_{i>k}\lambda_i$.

Theorem 3.1. *For every integer $s < k < t_s$, fixed-size determinantal sampling satisfies*

$$\frac{\mathbb{E}_k\mathcal{E}(S)}{T_k} \le \frac{k}{2(k-s)}\left(1+\sqrt{1+\frac{4(k-s)\lambda_{s+1}}{T_k}}\right) \tag{3}$$

$$\le \frac{k}{2(k-s)}\left(1+\sqrt{1+\frac{4(k-s)}{t_s-k}}\right) \tag{4}$$

$$\le \frac{k}{k-s}\sqrt{1+\frac{2(k-s)}{t_s-k}}. \tag{5}$$

No auxiliary ε, lower bound on $k-s$, or concentration-loss multiplier is required.

Proof. Write $\ell = k - s$, $\beta = \lambda_{s+1}$, and $T = T_k$. Since the eigenvalues are decreasing,

$$\mu(\alpha) \le s + \frac{\ell\beta}{\beta+\alpha} + \frac{T}{\alpha}.$$

Choose

$$\alpha = \frac{T + \sqrt{T^2 + 4\ell\beta T}}{2\ell}.$$

It solves $\ell\alpha^2 = T(\alpha+\beta)$, so the preceding upper bound on $\mu(\alpha)$ equals $s+\ell = k$. Corollary 2.2 gives $F_k \le k\alpha$, which is (3).

Moreover,

$$T = \beta r_s - \sum_{i=s+1}^{k} \lambda_i \ge \beta(r_s - \ell) = \beta(t_s - k) > 0.$$

This proves (4). Finally,

$$\frac{1+\sqrt{1+4x}}{2} \le \sqrt{1+2x} \qquad (x \ge 0)$$

follows by squaring and using $\sqrt{1+4x} \le 1 + 2x$. Substituting $x = (k-s)/(t_s - k)$ proves (5). $\square$

The final expression in (5) is the function $\Phi_s(k)$ appearing in [1, Theorem 1]. Theorem 3.1 therefore yields the loss-free version contemplated there, over the full range $s < k < t_s$, with the sharper intermediate bounds (3) and (4).

For completeness, the classical worst-case bound $F_k \le (k+1)T_k$ follows directly from $e_{k+1} \le T_k e_k$: every $(k+1)$-set contains an index greater than k, and expanding $T_k e_k$ counts its product at least once. Thus one may take the minimum of $k+1$ and any applicable bound above.

4 Scope and verification

The polynomial theorem concerns positive numbers; a rank-deficient matrix is handled by using only its positive eigenvalues, as in Section 2. Fixed-size determinantal sampling is asserted only for sizes at most the rank. The endpoint $k = r$ has zero residual and needs no relative-error ratio.

The accompanying script `verification/verify_volume_sampling.py` checks the second-difference identity symbolically, checks the polynomial inequalities in exact rational arithmetic, and checks the determinantal expectation identities on small matrices. These checks supplement, rather than replace, the proofs above.

References

[1] M. Dereziński, R. Khanna, and M. W. Mahoney. Improved guarantees and a multiple-descent curve for Column Subset Selection and the Nyström method. *arXiv:2002.09073*, version 3, 18 December 2020. `https://arxiv.org/abs/2002.09073`.